
NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
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 REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

CEA analysis performed on Thu 01-Jan-2026 12:51:49

Problem Type: "Rocket" (Infinite Area Combustor)

prob case = kaw00_____8420 ro equilibrium

Pressure (1 value):

p,psia= 120

Chamber/Exit Pressure Ratio (1 value):

pi/p= 12000

Oxidizer/Fuel Wt. ratio (1 value):

o/f = 2.2

You selected the following fuels and oxidizers:

reac

fuel C2H8N2(L),UDMH wt%= 50.0000

fuel N2H4 wt%= 50.0000

oxid N2O4 wt%=100.0000

You selected these options for output:

short version of output

output short

Proportions of any products will be expressed as Mass Fractions.

output massf

Heat will be expressed as siunits

output siunits

Plot parameters:

output plot isp

Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInputFile.cgi

IMPORTANT: The following line is the end of your CEA input file!
 end

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.500000	48900.000	298.150
FUEL	N2H4	0.500000	0.000	0.000
OXIDANT	N2O4	1.000000	0.000	0.000

O/F= 2.20000 %FUEL= 31.250000 R,EQ.RATIO= 1.022196 PHI,EQ.RATIO= 1.022196

CHAMBER THROAT EXIT

Pinf/P	1.0000	1.7262	12000.0
P, BAR	8.2737	4.7931	0.00069
T, K	3120.69	2971.85	784.54
RHO, KG/CU M	7.2851-1	4.4883-1	2.6228-4
H, KJ/KG	127.14	-474.11	-6076.82
U, KJ/KG	-1008.57	-1542.01	-6339.70
G, KJ/KG	-36928.4	-35762.3	-15392.5
S, KJ/(KG)(K)	11.8742	11.8742	11.8742
M, (1/n)	22.847	23.138	24.814
(dLV/dLP)t	-1.02885	-1.02486	-1.00000
(dLV/dLT)p	1.5858	1.5321	1.0000
Cp, KJ/(KG)(K)	6.4205	6.1670	1.5135
GAMMAS	1.1283	1.1260	1.2843
SON VEL,M/SEC	1132.0	1096.6	581.0
MACH NUMBER	0.000	1.000	6.062

PERFORMANCE PARAMETERS

Ae/At	1.0000	532.72
CSTAR, M/SEC	1681.0	1681.0
CF	0.6523	2.0954
Ivac, M/SEC	2070.4	3597.1
Isp, M/SEC	1096.6	3522.5

MASS FRACTIONS

*CO	0.07433	0.06670	0.00099
*CO2	0.11205	0.12404	0.22729
*H	0.00078	0.00061	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00006	0.00004	0.00000
*H2	0.00451	0.00393	0.00127
H2O	0.29557	0.30561	0.35172
H2O2	0.00001	0.00000	0.00000
*NO	0.01731	0.01385	0.00000
NO2	0.00002	0.00001	0.00000
*N2	0.41064	0.41226	0.41874
*O	0.00691	0.00523	0.00000
*OH	0.03800	0.03166	0.00000
*O2	0.03979	0.03604	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS